Supporting Information

Dynamically Heterogeneous Mesoscale Arrest in a Protein Polymer with Reversible β-Sheet-like Order

Doga Ozgulbas[1], Miaoqi Chu[2], Alex Lavens[1], Nicholas Marks[1], Ming Du[2], Xiaobing Zuo[2], Michael Irvin[1], Heng Ma[1], Soenke Seifert[2], Byeongdu Lee[2], Rafael Vescovi[1], Ian Foster[1], Victor Mateevitsi[3], Don Jensen Jr.[2], Chase Akins[4], Arvind Ramanathan[1,*], Gyorgy Babnigg[4,*], and Qingteng Zhang[2,*]

[1]Data Science and Learning Division, Argonne National Laboratory, Lemont, Illinois 60439, United States
[2]X-ray Science Division, Argonne National Laboratory, Lemont, Illinois 60439, United States
[3]Argonne Leadership Computing Facility, Argonne National Laboratory, Lemont, Illinois 60439, United States
[4]Biosciences Division, Argonne National Laboratory, Lemont, Illinois 60439, United States

*ramanathana@anl.gov; gbabnigg@anl.gov; qzhang234@anl.gov

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1 Recombinant Synthesis and Purification of $(VPAVG)_{30}$

1.1 Materials

Shrimp alkaline phosphatase, PfuUltra II Fusion HS DNA polymerase, GoTaq Green Master Mix, and T4 DNA ligase with PEG4000 additive were purchased from Thermo Fisher Scientific. Exonuclease I was purchased from New England Biolabs. CircLigase ssDNA ligase was purchased from Lucigen, and KOD Hot Start DNA polymerase was purchased from Sigma-Aldrich. PCR products and plasmids were purified using Monarch PCR and DNA Cleanup kits and Monarch Plasmid Miniprep columns (New England Biolabs). DNA was recovered from agarose gels using the Monarch DNA Gel Extraction kit (New England Biolabs). Oligonucleotides and the 5′-phosphorylated synthetic template used for rolling-circle amplification were obtained from Integrated DNA Technologies. *E. coli* BL21(DE3)-Gold cells were obtained from Stratagene.

1.2 Gene construction

The ELP gene was prepared by overlap-extension rolling-circle amplification (OERCA).[1] A single-stranded DNA template encoding five VPAVG repeats was circularized in a 20 µL reaction containing 10 pmol template, 2 µL of 10× CircLigase buffer, 1 µL of 1 mmol L^{-1} ATP, 1 µL of 50 mmol L^{-1} $MnCl_2$, and 100 U CircLigase. The reaction was incubated for 1 h at 60 °C, heat-inactivated for 10 min at 80 °C, purified, and treated with exonuclease I for 1 h at 37 °C to remove unreacted linear template.

OERCA was performed with 300 ng circular template, 1.5 µL each of 50 µmol L^{-1} primers A1 and A2 (Figure S1a), 2 µL of 70% GC dNTP mix, 2 µL of 25 mmol L^{-1} $MgSO_4$, 10 µL of 10× PfuUltra II buffer, 5 µL DMSO, and 2 µL PfuUltra II polymerase in a total volume of 100 µL. Cycling consisted of 2 min at 95 °C; 30 cycles of 20 s at 95 °C, 20 s at 54 °C, and 90 s at 72 °C; and a final 10 min at 72 °C. Products above approximately 1.2 kb were isolated from a 0.8% agarose gel.

Ligation-independent cloning adapters were introduced by PCR using primers B1 and B2 (Figure S1a), and the products were inserted into the pMCSG53 vector.[2, 3] Constructs were transformed into *E. coli* BL21(DE3)-Gold, screened for soluble expression, and sequence verified by Sanger sequencing. The

selected construct encodes the amino-acid sequence `MHHHHHHSSGVDLGTENLYFQSNA(VPAVG)30`, comprising a cleavable N-terminal affinity/fusion sequence followed by 30 VPAVG repeats. That leader was kept rather than cleaved, because its single tyrosine is the only chromophore in the chain and the A_{280} concentration measurement depends on it. Every measurement reported here is therefore on the tagged construct, and the leader accounts for 24 of the 174 residues and holds all eight charged residues.

1.3 Protein expression and purification

Fresh transformants were grown in Terrific Broth containing 10 mmol L^{-1} $MgSO_4$ and 300 µg mL^{-1} ampicillin at 37 °C to an optical density of 0.6–0.8 at 600 nm. Cultures were cooled to 18 °C, induced with 0.2 mmol L^{-1} IPTG, and grown overnight. Cells were harvested at 5000 × g, resuspended in ice-cold lysis buffer (500 mmol L^{-1} NaCl, 5 vol% glycerol, 50 mmol L^{-1} 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES) at pH 8.0, 20 mmol L^{-1} imidazole, and 10 mmol L^{-1} 2-mercaptoethanol), treated with 1 mg mL^{-1} lysozyme for 10 min, sonicated, and clarified at 29,000 × g.[4]

The supernatant was applied to 5 mL IMAC Sepharose High Performance resin, washed with lysis buffer, and eluted with lysis buffer containing 500 mmol L^{-1} imidazole. Purification fractions were monitored by SDS-PAGE. The purified protein was exchanged into 20 mmol L^{-1} HEPES at pH 8.0, 250 mmol L^{-1} NaCl, and 2 mmol L^{-1} dithiothreitol (DTT) and concentrated using a 10 kDa molecular-weight-cutoff centrifugal concentrator. Protein concentration was determined by direct A_{280} measurement on a NanoDrop spectrophotometer using the sequence-specific molar extinction coefficient applied during sample preparation (calculated molecular weight 15,461 Da and ε_{280} = 1490 M^{-1} cm^{-1}, both for the tagged construct). The final concentration was approximately 100 mg mL^{-1}; samples were stored at 4 °C, and freshly prepared protein was used for the X-ray measurements. The expression, solubility, and purity of the protein were evaluated by SDS-PAGE (Figure S1b).

2 Sample Environments and X-ray Measurement Setup

2.1 SA-XPCS at 8-ID-I

Small-angle X-ray photon correlation spectroscopy (SA-XPCS) was performed at APS 8-ID-I. Approximately 100 mg mL^{-1} $(VPAVG)_{30}$ in the same buffer was loaded into an aluminum cell containing a 3 mm-diameter, 2 mm-thick liquid chamber sealed by 0.127 mm polycarbonate windows. The cell was inserted into an anodized-aluminum holder in one temperature zone of the Quantum Northwest 3×3 multi-sample temperature control unit, the same stage used in Ref. 5. Each zone is driven by a Peltier element and a resistive heater, and a calibrated resistance-temperature detector (RTD) embedded adjacent to the Peltier stage supplies the controller feedback and readout. The aluminum holder and aluminum cell provided direct conductive thermal coupling. A partially coherent 10.94 keV X-ray beam with 0.03% FWHM bandwidth and incident flux 1.2×10^{10} photons s^{-1} was focused to approximately 10×10 μm^2. Coherent scattering was recorded 7.8 m downstream with a Rigaku XSPA-500k photon-counting detector comprising 1024×512 pixels with a 76 µm pixel size and a 320 µm-thick silicon sensor. The detector was operated in 2-bit mode at a 50 kHz frame rate; each acquisition contained 100,000 frames, corresponding to 20 µs per frame and 2 s total duration. The multi-tau delay list is built from the recorded 20 µs frame time, and the longest accessible delay is 1.638 s.[6]

Every 8-ID-I measurement reported here follows the same acquisition pattern. Single 2 s acquisitions were taken one after another, each at a previously unexposed position on a grid of sample positions, at a

cadence of roughly 6 s per acquisition once stage motion and readout are included. A single acquisition is therefore a snapshot of a fresh piece of sample at a known time, and a series steps through increasing time at that cadence. SAXS profiles were obtained by temporal and azimuthal averaging. Every reported quantity is a group average rather than a single acquisition: consecutive acquisitions were averaged in contiguous run ranges after the two outlier cuts of Section 5.3, and the quoted time is the acquisition time of the first run in the range. The thermal protocol is given with its results in Section 6.1.

3 8-ID-I Absolute Scattering Cross Section

3.1 Absolute scattering cross section

For the 8-ID-I data, the differential scattering cross section was calculated as

$$I(Q) = A_s I_s(Q) - A_b I_b(Q), \tag{S1}$$

where $I_s(Q)$ and $I_b(Q)$ are the temporal-mean detector intensities from the sample and from its background, per pixel and per frame. The background is a buffer where one was measured and the empty cell of Section 6.2 otherwise. The conversion factors were

$$A_j = \frac{N_\mathrm{f}}{F_j T_j t_\mathrm{exp} L \Delta\Omega}, \qquad j \in \{s, b\}, \tag{S2}$$

where F_j is the incident photon flux, T_j is the transmission, $N_\mathrm{f} = 100{,}000$ is the number of frames, $t_\mathrm{exp} = 2\,\mathrm{s}$ is the total exposure time, so that $t_\mathrm{exp}/N_\mathrm{f}$ is the 20 µs frame time, $L = 2\,\mathrm{mm}$ is the sample thickness, and $\Delta\Omega = (d_\mathrm{pix}/D)^2$ is the solid angle per detector pixel.[7] Lengths are carried in millimeters throughout, so $A_j I_j$ is in mm^{-1}, and that is the quantity plotted in Figures 2a and S4–S7: no further scaling of any kind is applied between the calibration and the figure.

3.2 Transmission and incident-flux calibration

Sample transmission was obtained from ion chambers immediately upstream and downstream of the sample. Across the attenuation series, the transmission of the intervening air path was independent of beam intensity within uncertainty, with a mean value of 0.8679 ± 0.0009 (Figure S2a; mean ± standard deviation). The upstream ion-chamber signal was converted to incident photon flux by calibration against a PIN diode positioned in the flight path upstream of the beamstop. The measured photon flux varied linearly with the upstream ion-chamber readout (Figure S2b).

4 8-ID-I SA-XPCS Correlation Analysis

4.1 Correlation functions and the fitted model

The intensity autocorrelation function was calculated with a multi-tau algorithm,

$$g_2(Q, \Delta t) = \frac{\langle I(Q,t) I(Q, t+\Delta t)\rangle_t}{\langle I(Q,t)\rangle_t^2}. \tag{S3}$$

The implementation is the 8-ID-I reduction workflow; its numerical treatment of the multi-tau scheme at these frame rates is described in detail elsewhere.[6, 8]

The reduction produced 27 azimuthally averaged dynamic Q bins of nominal width 9.9×10^{-4} Å^{-1} spanning 0.003 59–0.029 37 Å^{-1}, and 61 delay times from 20 µs to 1.64 s. The five lowest bins, centered at $Q = 0.00359$, 0.00458, 0.00557, 0.00657 and 0.00756 Å^{-1} and corresponding to $2\pi/Q$ of approximately 175, 137, 113, 96 and 83 nm, are analyzed here. The restriction is a signal-to-noise limit: the scattered intensity falls by more than an order of magnitude between the lowest bin and 0.00756 Å^{-1}, so above that bin the correlation function is dominated by counting noise.

The measured g_2 carry two decays separated by several decades in delay, and were parameterized through a normalized intermediate scattering function g_1 built from two stretched exponentials,

$$\begin{aligned} g_2(Q,\Delta t,t_w) &= 1 + \beta\,|g_1(Q,\Delta t,t_w)|^2\,, \\ g_1(Q,\Delta t,t_w) &= f(Q,t_w)\exp\!\left[-\left(D_0(t_w)\,Q^2\,\Delta t\right)^{p_{\mathrm{fast}}}\right] \\ &\quad + [1-f(Q,t_w)]\exp\left[-\left(\frac{\Delta t}{\tau_{\mathrm{slow}}(Q,t_w)}\right)^{p_{\mathrm{slow}}(t_w)}\right]. \end{aligned} \tag{S4}$$

The first line is the Siegert relation. The two-mode form on the second line is the same one used for thermoreversible nanoparticle gels, where the two-step decay is fitted as a linear sum of two stretched exponential modes with complementary amplitudes a and $1-a$.[9] The same two-mode decomposition, with a fast decay and a slow stretched decay whose amplitude sets the intermediate plateau, describes ergodicity breaking in aging nanoclay suspensions.[10] Here g_1 is the normalized intermediate scattering function, f is the fitted fraction of fast dynamics and $1-f$ its complement. β is the instrumental speckle contrast, set by the transverse coherence of the beam and by the ratio of speckle size to pixel size. It is a property of the beamline and the detector rather than of the sample, is measured on a static reference, and is held fixed at 0.141 throughout (Section 6.2). τ_{slow} is free at each Q and each t_w. The fast relaxation time is not free, being tied to the diffusive form $\tau_{\mathrm{fast}}(Q,t_w) = 1/[D_0(t_w)Q^2]$, so one coefficient replaces five free times at each waiting time. The Kohlrausch–Williams–Watts exponents carry no Q argument because they are shared across the five bins. p_{fast} is free at each waiting time rather than shared along a hold: these holds run from a mobile state into arrest, p_{fast} falls from about 1 to about 0.2 as they do, and a single shared value is rejected by an F test at $p < 10^{-16}$ in every loading that arrests. The decomposition is phenomenological: it describes the observed first decay and persistent plateau without assigning either mode to a unique molecular mechanism.

4.2 Global fitting and uncertainties

Equation S4 was fitted to all five Q bins of one averaged group at once. f and τ_{slow} are free in each bin; D_0, p_{fast} and p_{slow} are single values shared across the bins of that group. Groups are fitted independently of one another, so nothing is shared along a hold. Fits minimized the error-weighted residuals $[g_{2,\mathrm{model}} - g_{2,\mathrm{data}}]/\sigma_{g_2}$ with `scipy.optimize.least_squares`, using the pointwise g_2 uncertainties stored in the reduced files, so each delay enters with weight $1/\sigma_{g_2}^2$ and the noisy short delays count for little. Bounds were wide ($0 \le f \le 1$, $1 \le \tau_{\mathrm{slow}} \le 10^4$ s, $0.2 \le p \le 3$, $10^2 \le D_0 \le 10^{10}$ Å^2 s^{-1}). Each fit is started from the minimum of a p_{fast} profile so that it cannot settle in a solution where the fast term imitates the slow one. Reduced χ^2 runs from 0.90 to 3.34 over the four loadings, and a group is kept only if it stays below 10. Starting values and the remaining numerical settings are in the code (Section 7).

Parameter uncertainties are the square roots of the diagonal of the covariance matrix, estimated as the Moore–Penrose pseudo-inverse of $J^{\mathrm{T}}J$ with J the Jacobian of the error-weighted residuals at the solution; the pseudo-inverse is used so that a parameter refining to a bound does not leave the uncertainties of the remaining, well-constrained parameters undefined. The fast fraction reaches its upper bound of one in 14

of the 170 bin fits, all of them in still-mobile groups of the 28.00 and 28.25 °C holds; its smallest fitted value anywhere is 0.07. These are the error bars plotted in Figure 2c and in Figures S4–S7.

The slow component remains only partially relaxed during the 2 s acquisition and frequently yields $\tau_{\mathrm{slow}} > 2$ s, so it is an effective extrapolation rather than a sampled terminal relaxation time. Conclusions about arrest therefore rest on directly observed features: the decreasing amplitude of the first decay, the rising plateau, and the systematic decrease in f. XPCS does not determine where the arrested regions are, $1 - f$ is not a geometric volume fraction because scattering contributions are weighted by structure and contrast, and because the measurement does not reach infinite delay it is not the formal nonergodicity factor either. Panels (d)–(g) of Figures S4–S7 give the correlation functions and fits at the four additional Q values: the same two-step shape and the same monotonic loss of the fast step appear at every measured Q, so the trend of Figure 2c belongs to the whole 83–175 nm range and not to a single bin.

4.3 Stretching exponents and the fast-mode scaling

Both stretching exponents stay below unity. p_{fast} runs from 0.55 to 1.00 while a hold is still mobile and from 0.20 to 0.51 once it has arrested, and p_{slow} is free as well. That the breadth of the fast relaxation grows as the material locks up is itself a measurement rather than a modelling choice: a single p_{fast} shared along a hold is rejected by an F test at $p < 10^{-140}$ in each of the three loadings that arrest, while in the 28.00 °C loading, which does not arrest, freeing it moves the reduced χ^2 only from 1.21 to 1.19. Stretched decays of this kind indicate a broad distribution of relaxation times rather than a single process, and are characteristic of protein and colloidal gels,[9, 11] although the direction of the departure from $p = 1$ varies between systems: the egg-white gel of Ref. 11 relaxes with compressed rather than stretched exponentials, the signature of stress-driven rather than diffusive relaxation.

The fast relaxation occurs on approximately 10^{-4} to 10^{-3} s and becomes faster with increasing Q. Equation S4 imposes the diffusive scaling $\tau_{\mathrm{fast}} \propto Q^{-2}$ rather than fitting it, and a stretched g_2 remains compatible with diffusion provided p does not depend on Q, which holds here by construction. Releasing the constraint, so that τ_{fast} is free in every bin, returns a median Q exponent of -2.5 over the 32 groups whose fast amplitude at the lowest Q exceeds 0.15, with 80% of them between -2.6 and -1.3. That release is also the general two-mode form of Ref. 9, in which the fast time is free in every bin rather than carried by one coefficient. Refitting every group that way moves f at the lowest Q by 0.006 at the median, by 0.005 or less in the groups where the fast mode has nearly gone, and by at most 0.16 in a single group, and it moves the scaling temperature of Figure 3 from 0.224 to 0.220 K, a shift of 1.7% against a bin-to-bin spread of 5%. Those fits are seeded as every other fit here is, from the minimum of a p_{fast} profile. That matters, because with p_{fast} free the two-mode form has local minima in which the fast term imitates the slow one: started instead from the bare default of 0.5, two of the 34 groups settle in one of them at reduced χ^2 of 28 and 93, against 2.6 and 9.2 seeded, and the shift turns the other way, to +0.9%. Both are small against the spread across Q. The collapse therefore does not rest on the diffusive constraint. Where it is not, that is in the later groups of the holds that arrest furthest, the exponent collapses toward zero because τ_{fast} stops being constrained once the fast decay has gone, not because the relaxation changes character. Those exponents bracket the imposed value of -2 but sit systematically steeper than it, so the diffusive form is reported here as an imposed approximation that the data are consistent with, and not as a fitted result. What is read from it is the fitted coefficient D_0 of Section 6.3, and the conclusions of this work rest on f rather than on D_0: f is set by the height of the plateau and stays measured to the end of every hold, whereas any fast-mode time scale needs a resolved decay. Freezing p_{fast} at 0.5 for every group, against fitted values spanning 0.20 to 1.00, moves f at the

lowest Q by at most 0.07 on a range of 0.07 to 0.99 and the scaling temperature of Figure 3 by 3%, so neither result depends on how the stretching is treated.

5 Radiation-Damage Mitigation and Controls

5.1 Fresh sample positions

The grid of positions referred to in Section 2.1 was Cartesian, 100×100 points in the aluminum cell, with adjacent points separated by at least one beam diameter. Because no position was visited twice, cumulative dose cannot track the elapsed time of the isothermal experiment, so the trend in the fitted fast fraction cannot be a dose artifact.

5.2 Flux-dependence control

In addition, to show that there is no significant radiation-induced artifact, an independent dose-rate test was performed after the APS Upgrade using a $20\,\mathrm{mg\,mL^{-1}}$ $(\mathrm{VPAVG})_{30}$ sample. Every acquisition was taken at a stage-RTD readout of 33.0 °C, above the transition temperature, so the control probed the assembled high-temperature state rather than the low-temperature dispersed one. Attenuation selected incident fluxes from 2.7×10^{11} to 4.9×10^{11} photons $\mathrm{s^{-1}}$, approximately 23–41 times the flux used for the main experiment. The SAXS line shapes and g_2 decays showed no systematic flux dependence (Figure S3). That control sample is five times more dilute than the $100\,\mathrm{mg\,mL^{-1}}$ material of the main experiment, and it was measured with the post-Upgrade beam. The main series is protected instead by the fresh position used for every acquisition (Section 5.1).

5.3 Outlier rejection

Two cuts are applied to every group in this work on identical terms, with no per-group list. The first compares each acquisition with the group on the shape of g_2, g_2 uncertainty and $I(Q)$ by cross-correlation, and drops the lowest fifth percentile of each similarity score; being a percentile rather than a threshold, it removes 10 to 15% of every group whatever the data look like. The second is robust rather than fractional and catches what the first does not: a large object drifting through the 10×10 $\mu\mathrm{m}^2$ beam during a minority of acquisitions raises $I(Q)$ in the lowest few Q bins alone, which barely moves a whole-profile shape score. For each acquisition the mean of $I(Q)$ over $0.004 < Q < 0.008$ $\mathrm{Å^{-1}}$ is compared with the group median, and the acquisition is rejected if it exceeds that median by more than three robust standard deviations, the scale being $1.4826\times$ the median absolute deviation, both statistics iterated to convergence. The test is one-sided because a transient scatterer only adds intensity. The threshold is not tuned: the same acquisitions are selected for any value between 2.5 and 3 and for two other choices of band.

6 Temperature Dependence of Arrest

6.1 Measurements and reduction

The four loadings that carry the main text were measured at 8-ID-I in October 2021, with the beam energy, detector, detector distance and cell design of Section 2.1. They are the same construct as the

12-ID-B material, from a separate preparation. After a reference measurement at 6 °C, each loading was heated from 20 °C at 0.1 °C min^{-1} to its final temperature and held there: loading D8 at 28.00 °C, D6 at 28.25 °C, D7 at 28.50 °C and D9 at 28.75 °C. The temperatures are the readings of the stage RTD, and $t_w = 0$ is the first acquisition with the RTD at its target. Acquisitions of 100,000 frames ran at fresh positions through the ramp and then for 7100, 8500, 3900 and 2200 s of the hold, respectively.

The 2021 Data Exchange records were converted to the NeXus format with the incident- and transmitted-beam monitors and the ring current carried across unchanged, and the frames were correlated with the 8-ID-I reduction workflow of Section 4. Consecutive acquisitions were averaged in groups of 100, about 10 min each, with one group boundary at $t_w = 0$, through the same two outlier cuts that every other group in this work receives (Section 5.3). One detector pixel read a constant 1.5 photons per frame in the measurements of 17 October and is masked in the Q map the series is correlated against. It lay at $Q = 0.0214$ Å^{-1}, outside the five Q bins of the correlation analysis; masking it left every fitted quantity unchanged.

6.2 Absolute scale, background and contrast

The beamtime used the same ion chambers as the calibration of Section 3, so $I(Q)$ is put on that absolute scale, each group with its own monitor readings. The four loadings transmit 0.383 to 0.389. The background is a separate measurement of the sample cell, subtracted on the absolute scale exactly as a buffer would be. Six candidate measurements are available, four of them recorded on the same day as the loadings. They are identified here by what they transmit and what they scatter rather than by the labels entered at the beamline, which do not reliably distinguish a cell left empty from one holding buffer. Four of the six recorded a transmitted-beam monitor. One of those passes 0.94 of the beam and the other three pass 1.00, the same as the open path, so one has of order 6% of attenuating material in the beam and the others have none that the chambers resolve. None passes anything near 0.41, which is what a 2 mm column of buffer behind the same material would give, so none of them holds a liquid column of the sample thickness. The measurement that passes 0.94 is accordingly the only one of the six that carries scattering from material in the beam path, and that is what the subtraction needs. Fitted by the procedure below, the three that pass 1.00 need a coefficient between 3 and 5 and still leave the 6 °C references 7 to 28% from flat at best, against 3 to 5% for the one that passes 0.94, with the residual peaking near 0.008 Å^{-1}. Averaging the three together does not improve on the best single one, so what they lack is shape and not counting statistics. Whether any of these measurements held buffer changes little over this Q range: the absorption difference is taken up by the coefficient, and the scattering of a 2 mm water column is about 0.0016 mm^{-1}, some 6% of the plateau and flat in Q, so it shifts the level rather than the shape.

How much of that background to subtract is set by the 6 °C state, which is dissolved and whose profile should fall smoothly. Each reference is reduced to the median in each of nine logarithmic Q bins between 0.0036 and 0.0305 Å^{-1}, and two numbers describe the result: the rise, its lowest bin divided by its highest, and the largest fractional increase from one bin to the next going up in Q, which is zero when the profile never turns back up. The coefficient trades one against the other and nothing in these data escapes that trade. At $\alpha = 0.865$, the value that makes the references flattest, the rise is 1.0 but the profile dips 6 to 8% near 0.004 to 0.005 Å^{-1}, and a dip is a subtraction artefact rather than anything the sample does. Removing the dip altogether takes $\alpha = 0.60$, which leaves a two-fold climb toward low Q in its place. We use $\alpha = 0.80$ for all four loadings. The rise is then 1.26 to 1.45 and the profile turns back up by at most 4%, below the scatter of the individual points, and the 6 °C references sit at 0.026 to 0.028 mm^{-1} over

0.006 to 0.03 Å^{-1}.

Figures S4–S7 show $I(Q)$ from the lowest fitted Q bin outward. Below that bin the points sit on the edge of the beamstop shadow, where the background and the sample no longer share a profile. None of this beamtime's background measurements holds a buffer column, as the transmissions above show, so the subtraction cannot take windows, solvent and parasitic scattering off together the way a buffer measured in the same cell does. At 0.004 Å^{-1} the background removed is 2.4 to 3.0 times the 6 °C signal left behind, so subtracting enough to remove the instrumental tail would take the sample's own low-Q signal with it, and the 6 °C references keep a residual rise at every coefficient. A mismatch of the Q scale between the two was tested as the cause and rejected: flattening the residual that way takes a 6% stretch of the background Q axis and improves the agreement by half a percentage point. Averaged over 0.004 to 0.008 Å^{-1}, the 34 fitted groups stand 54 to 322 times the background, so this limit bears on the 6 °C reference alone.

Loading D8 was measured at 6 °C twice, before and after its hold, and Figure S4 shows the later measurement. The earlier one was the first measurement of the beamtime and is not a dissolved state: it carries a $Q^{-2.7}$ excess at low Q that falls to 4% of the plateau above 0.01 Å^{-1}, and it fits $\alpha = 1.15$ against 0.85 to 0.88 for every other reference. The later measurement is the same loading in the same cell, returned to 6 °C after the 28.00 °C hold; it is flat and fits $\alpha = 0.860$. The excess is therefore undissolved material in the earlier measurement rather than a property of the subtraction.

The speckle contrast was measured on a static reference recorded in the same beamtime, as the flat level of its correlation function through the Siegert relation that opens Equation S4. A static sample cannot relax, so each of its correlation functions is flat in delay and sits at $1 + \beta$. The value is *not* taken over the five bins the correlation analysis fits, where that reference reads 0.1226, 0.1323, 0.1355, 0.1372 and 0.1390 with increasing Q: the standard is scratched, and its static parasitic scattering depresses the apparent contrast most strongly at low Q. β is instead read from the plateau the same reference reaches between 0.0085 and 0.0165 Å^{-1}, where nine consecutive bins give 0.1379 to 0.1428 and their inverse-variance mean is $\beta = 0.141$, the single value used for every bin and every group. Using the fitted bins instead would understate β by 5%, which returns too large an f: the arrested fractions quoted below would rise from 0.93 to 0.96, while the ordering of f in Q and the scaling temperature of Figure 3 would be unchanged.

6.3 Fitting

Equation 2 of the main text was fitted to the five lowest Q bins, 0.00359 to 0.00756 Å^{-1} (175 to 83 nm), with β fixed. A group is fitted when its correlation function is no noisier than a fixed threshold: the median g_2 uncertainty at the lowest Q between 10^{-4} and 10^{-2} s may not exceed 0.016. Two choices follow from what the holds contain. First, p_{fast} is fitted in each group. The holds run from a largely mobile state, where p_{fast} is 0.55 to 1.00, into arrest, where it is 0.20 to 0.51, and a single value shared across a hold is rejected by an F test in every loading, at $p < 10^{-140}$ in the three that arrest. Each fit starts from the minimum of a p_{fast} profile, so that it cannot settle in a solution in which the fast term imitates the slow one. Second, the rate of the fast mode is reported as the fitted coefficient D_0 itself. No mean or median of the stretched decay is formed, because either one is dominated by the tail of the distribution and moves strongly with p_{fast}. Reduced χ^2 is 0.90 to 3.3. Early in arrest τ_{slow} reaches its upper bound of 10^4 s in 17 of 170 bin fits, where the slow mode does not decay within the acquisition; f is set by the height of the plateau and is unaffected.

Fixing p_{fast} would remove one parameter per group, and the data were tested against it. At $p_{\mathrm{fast}} = 1$,

where the fast mode is a single exponential, the mean reduced χ^2 of the 28.00 °C hold rises only from 1.18 to 1.41, so a mobile sample does relax exponentially. The three holds that arrest rise from 1.75, 1.89 and 1.64 to 8.27, 9.56 and 7.22, and the fitted curve crosses the data rather than following it. Intermediate fixed values do not recover the fit: averaged over all 34 groups, $p_{\mathrm{fast}} = 1$ gives 6.60, 0.7 gives 3.73 and 0.5 gives 2.44, against 2.21 for the best single value shared within each hold and 1.62 with p_{fast} fitted per group. The broadening of the fast relaxation is therefore a feature of the data in the arresting holds rather than a free parameter absorbing noise.

6.4 Results

Figures S4–S7 show the four loadings in the format of Figure 2 of the main text, with the correlation functions at the four further Q values in panels (d)–(g). Assembly begins before any target is reached. Averaged over $0.004 < Q < 0.008$ Å^{-1}, the scattered intensity rises on the ramp to 2.3 to 3.6 times its level at 20 to 22 °C by 27.5 °C, in every loading.

At 28.00 °C the sample assembles without arresting. The same low-Q intensity grows 22-fold over the hold, to 6.2 mm^{-1}, while f stays at 0.89 or above at every Q for 6500 s.

At the three higher temperatures the sample arrests, and the rate rises steeply with temperature. f at the lowest Q falls to 0.5 after 5810, 1930 and 540 s of hold at 28.25, 28.50 and 28.75 °C. Measured instead from the moment the RTD passed 28.00 °C, which removes the different ramp times, and taking each group at its midpoint, the times are 6240, 2520 and 1280 s, a factor of five over 0.5 °C. In every group in which arrest is under way ($f < 0.85$ at the lowest Q), f increases monotonically with Q: 7 of 7, 7 of 7 and 4 of 4 groups. Figure 3 puts f at the largest length scale for all four holds on one axis. At 28.50 °C the last group gives $f = 0.07$ at 175 nm and 0.16 at 83 nm.

The structure does not set the arrest. At 28.25 °C the first group in which arrest is under way ($t_w = 4140$ s) has an $I(Q)$ between 0.88 and 1.12 times the final 28.00 °C profile at every Q from 0.004 to 0.028 Å^{-1}. At 28.75 °C ($t_w = 6$ s) the ratio is 0.82 to 0.92. That 28.00 °C profile stays mobile. At 28.50 °C arrest sets in at 0.48 to 0.69 of it. Once under way, arrest proceeds while the low-Q intensity keeps rising, by a factor of 1.6 to 4.0 between the first group in which arrest is under way and the last.

The mobile population slows before arrest and then changes far less than the amplitude does. While f at the lowest Q stays at 0.7 or above, D_0 falls 2.7-, 3.8- and 3.4-fold at 28.00, 28.25 and 28.50 °C. Below 0.7 it varies by factors of 1.7, 1.9 and 3.0 at 28.25, 28.50 and 28.75 °C, while $1 - f$ at the lowest Q rises from 0.33, 0.46 and 0.53 to 0.86, 0.93 and 0.84.

7 Code, Data and AI-Assisted Development

7.1 Code and reduced-data availability

The reduced datasets and all analysis and figure-generation code used for this manuscript are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP`. The repository includes the reduced 12-ID-B SAXS/WAXS data, the averaged 8-ID-I XPCS HDF files for every group plotted, the per-acquisition temperature record of the thermal-cycling sequence, radiation-control data, absolute-intensity calibration inputs, and the scripts used to generate the main-text and Supporting Information figures. Every data-derived figure can be regenerated from the repository alone. Two further scripts document how the committed reduced files were produced from the raw beamline results.

`SAXS_12id/Read_12ID_SAWAXS.py` handles 12-ID-B, merging the SAXS and WAXS profiles into the committed CSV files. `SAXPCS_8id/average_ranges.py` handles 8-ID-I, applying both outlier-rejection passes to every group on the same terms and averaging each into the committed HDF files. Both read the beamline storage directly. The repository README maps each script and output file to the corresponding manuscript figure. The raw detector frames occupy 3.0 TB and the reprocessed per-acquisition results 51 GB, both held on Advanced Photon Source project storage inside the facility network. The 196 MB of averaged files committed here are what the figures are produced from. Raw data are available from the corresponding authors.

7.2 AI-assisted development

The authors used Claude Sonnet 5 and Opus 5 to assist with the manuscript preparation and code development. The Claude models were accessed through Argo, Argonne National Laboratory's internal generative AI interface, which provides Laboratory staff with access to large language models from multiple vendors, including Anthropic, OpenAI, and Google, for use in Laboratory research and operations.

Qingteng Zhang reviewed the code, reran the analyses, and confirmed the reported results. The code is public (Section 7.1), so every reported number and data-derived figure can be checked independently. All analysis choices, scientific interpretations, and manuscript content were determined by the authors, who take full responsibility for the publication.

(a)

A1 (forward)
5′-GTTCCAGCTGTTGGTG

A2 (reverse)
5′-ACCTACTGCTGGTACG

B1 (forward)
5′-TACTTCCAATCCAATGCCGTTCCAGCTGTTGGTGTGCCG

B2 (reverse)
5′-TTATCCACTTCCAATGTTAACCTACTGCTGGTACGCCGAC

$VPAVG_{30}$ OERCA single-stranded template
/5Phos/GTTCCAGCTGTTGGTGTGCCGGCCGTGGGCGTGCCTGC
GGTGGGCGTCCCGGCGGTCGGCGTACCAGCAGTAGGT

(b)

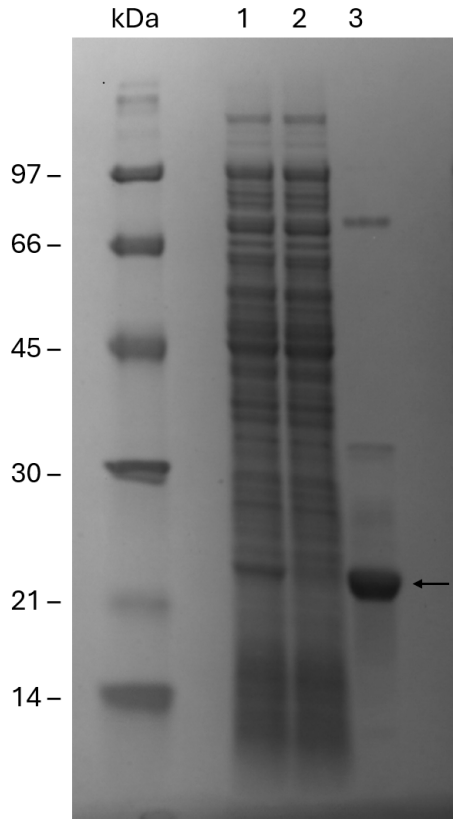


Figure S1: Construction and expression of the $(VPAVG)_{30}$ gene. (a) Oligonucleotide primers and the 5′-phosphorylated single-stranded OERCA template used to construct the ELP gene. (b) SDS-PAGE analysis of expression and purification. Samples were resolved by SDS-PAGE and stained to evaluate expression, solubility, and purity of the heterologously expressed protein. The molecular-weight marker lane is at the left, with prominent bands labeled in kilodaltons (kDa). Lanes 1–3 are the total soluble fraction, the flow-through fraction after IMAC, and the eluted fraction, respectively. The arrow marks the eluted $(VPAVG)_{30}$ construct, whose apparent mobility is lower than that expected from its calculated mass of 15,461 Da.

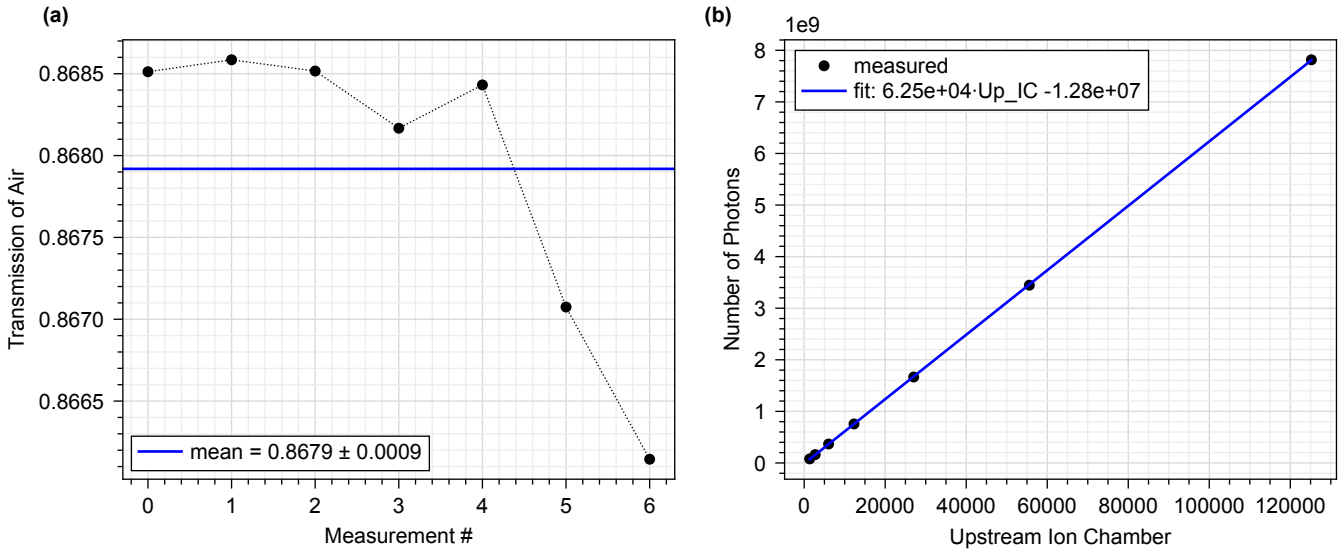


Figure S2: Transmission and photon-flux calibration for conversion to absolute scattering cross section. (a) Air-path transmission across seven measurements in the attenuation series; the line marks the mean, 0.8679 ± 0.0009 (standard deviation). (b) Photon counts recorded by a calibrated PIN diode as a function of the upstream ion-chamber readout (both in the scaled detector units of the calibration series); the line is the linear calibration fit, from which the incident flux follows as $F = (A \cdot \mathrm{Up_IC})/t_{\mathrm{exp}} + B$.

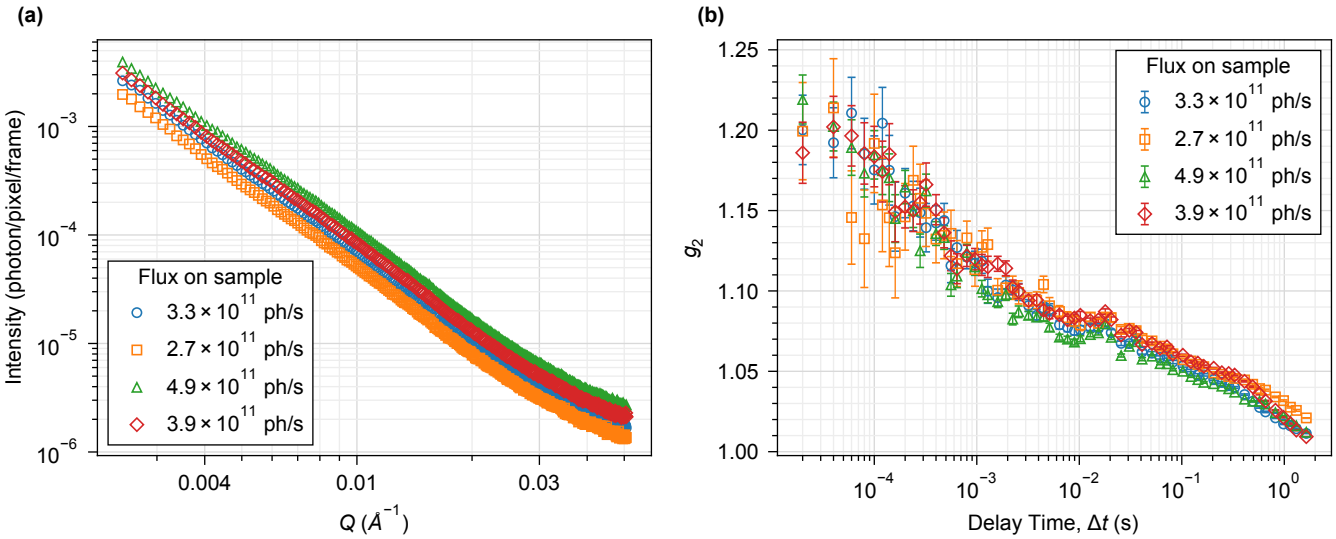


Figure S3: Flux-dependence control for SAXS and SA-XPCS. (a) SAXS profiles and (b) intensity autocorrelation functions from a 20 mg mL^{-1} $(\mathrm{VPAVG})_{30}$ sample at incident fluxes of 2.7×10^{11}–4.9×10^{11} photons s^{-1}. The Q range shown in (a), 0.0024–0.051 Å^{-1}, covers the Q range of Figure 2a of the main text. Intensities in (a) are detector counts per pixel per frame, not on the absolute scale of Figure 2a, so the vertical spacing of the four curves is the incident flux itself: their mean intensities stand in the ratios 1.00, 1.29, 1.58 and 2.01 against incident-monitor ratios of 1.00, 1.24, 1.47 and 1.82. The line shapes nonetheless stay parallel across that range. Panel (b) shows the lowest Q bin of this control configuration, $Q = 0.00304$ Å^{-1} ($2\pi/Q \approx 206$ nm). The legend orders the four conditions by nominal attenuator setting, which is not monotonic in flux. No systematic flux-dependent change is resolved over this range, which substantially exceeds the flux used in the main experiment.

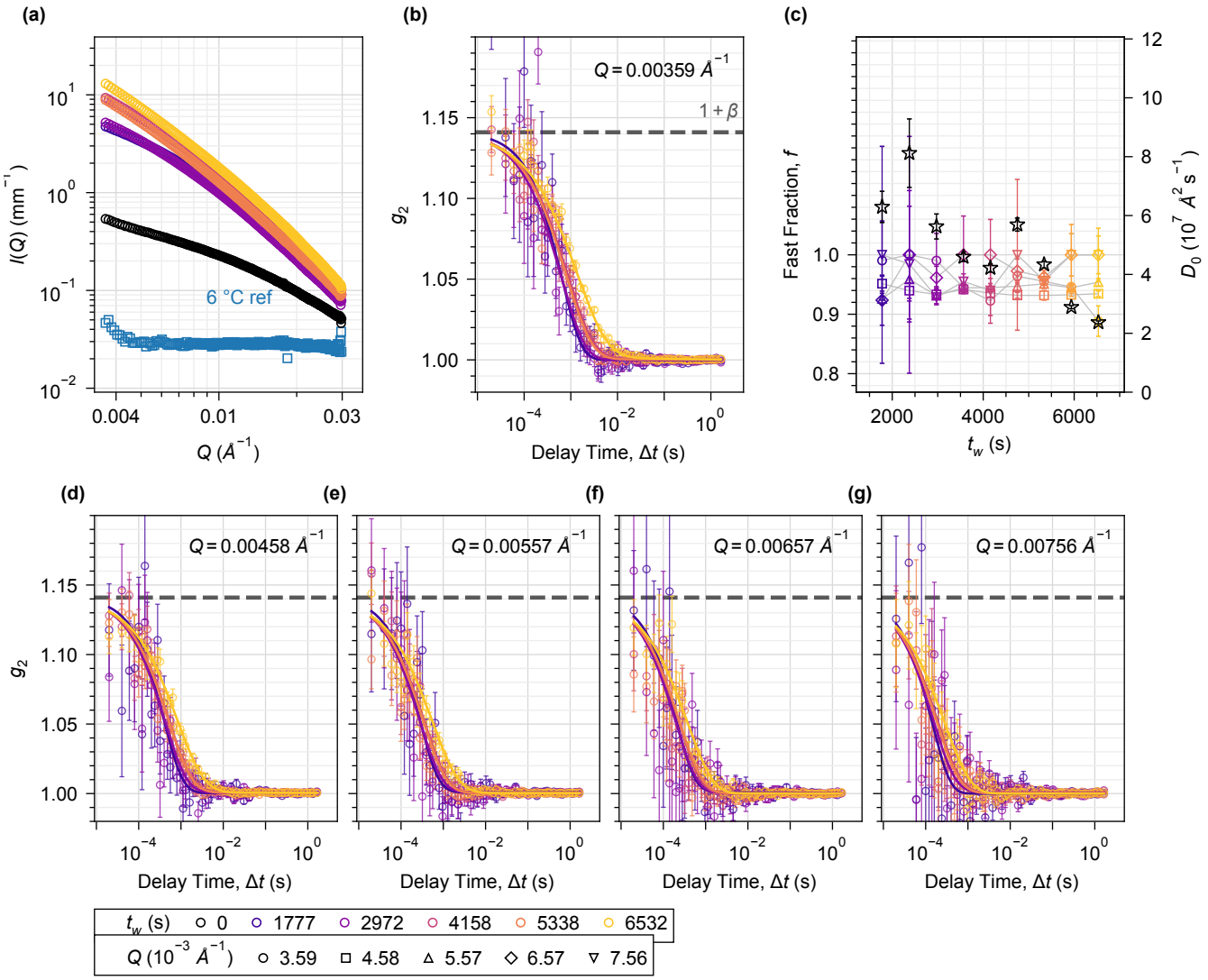


Figure S4: Loading D8 held at 28.00 °C, where it assembles without arresting: the low-Q intensity grows throughout the hold while f stays at 0.89 or above at every Q. (a) Background-subtracted SAXS on the absolute scale of Figure 2a of the main text: the 6 °C profile of loading D8 (blue squares), measured after the hold rather than before it (Section 6.1), the first group at 28.00 °C and later groups. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 of the main text at $\beta = 0.141$, p_{fast} free in each group; the dashed line in (b) and (d)–(g) is $1 + \beta$, the contrast of Section 6.2. (c) Left axis, open coloured markers: the fast fraction f at the five Q values, joined by grey guide lines. Right axis, open black stars: the fitted diffusion coefficient of the fast mode, D_0. (d)–(g) g_2 and fits at the four further Q values. Colors denote t_w, keyed along the bottom; (a), (b) and (d)–(g) show every second group, and the black profile at 0 s is too noisy to fit. Error bars are one-standard-deviation fit uncertainties; Section 6 gives the analysis.

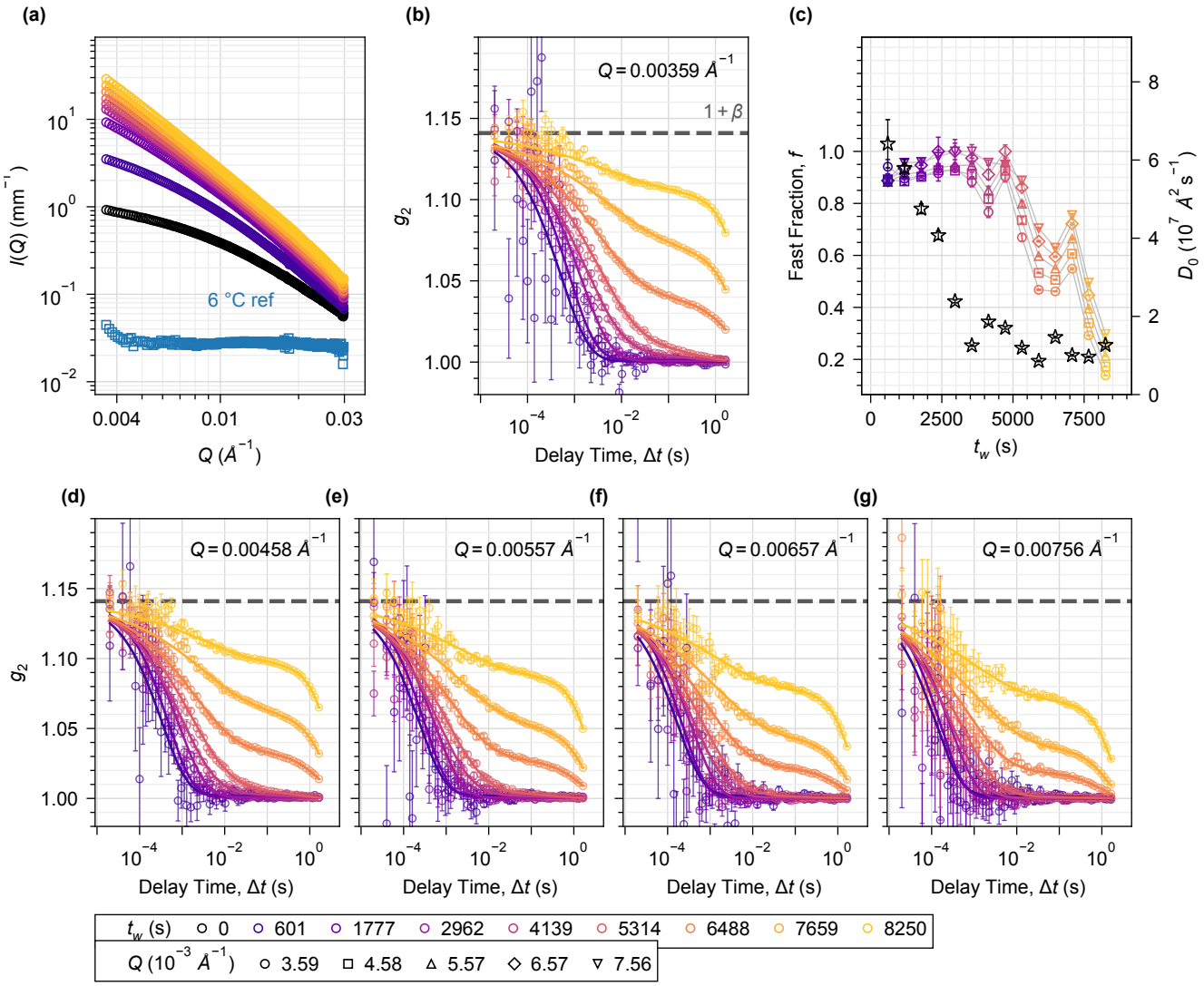


Figure S5: Loading D6 held at 28.25 °C. After about 4000 s the sample arrests, f at the lowest Q falling to 0.14 by 8250 s and increasing with Q at every t_w, while D_0 has already fallen before arrest and then levels off. (a) Background-subtracted SAXS on the absolute scale of Figure 2a of the main text: the 6 °C reference of loading D6 (blue squares), the first group at 28.25 °C and later groups. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 of the main text at $\beta = 0.141$, p_{fast} free in each group; the dashed line in (b) and (d)–(g) is $1 + \beta$, the contrast of Section 6.2. (c) Left axis, open coloured markers: the fast fraction f at the five Q values, joined by grey guide lines. Right axis, open black stars: the fitted diffusion coefficient of the fast mode, D_0. (d)–(g) g_2 and fits at the four further Q values. Colors denote t_w, keyed along the bottom; (a), (b) and (d)–(g) show every second group, and the black profile at 0 s is too noisy to fit. Error bars are one-standard-deviation fit uncertainties; Section 6 gives the analysis.

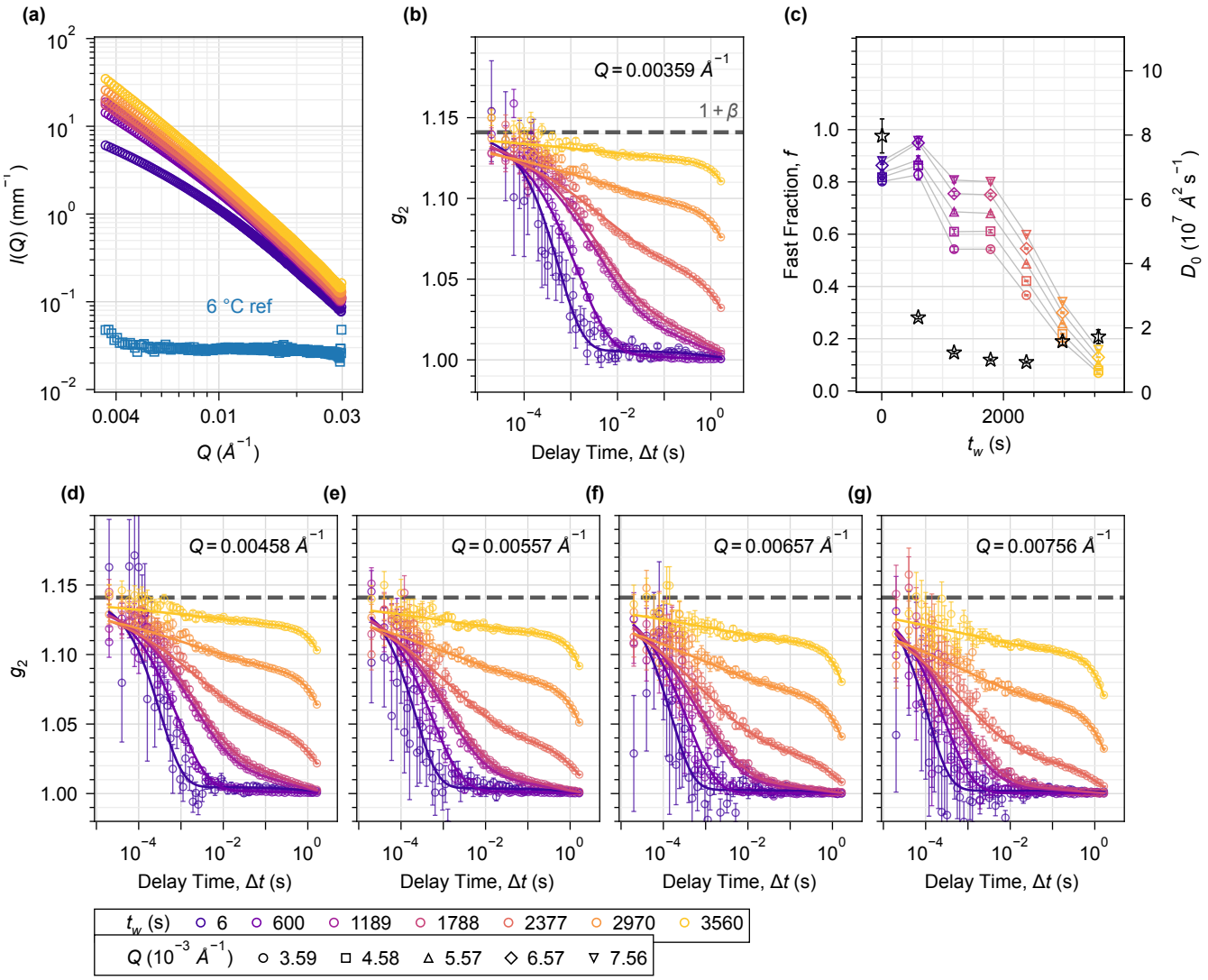


Figure S6: Loading D7 held at 28.50 °C. f at the lowest Q falls from 0.80 to 0.07 within 3600 s, increasing with Q at every t_w. (a) Background-subtracted SAXS on the absolute scale of Figure 2a of the main text: the 6 °C reference of loading D7 (blue squares), the first group at 28.50 °C and later groups. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 of the main text at $\beta = 0.141$, p_{fast} free in each group; the dashed line in (b) and (d)–(g) is $1 + \beta$, the contrast of Section 6.2. (c) Left axis, open coloured markers: the fast fraction f at the five Q values, joined by grey guide lines. Right axis, open black stars: the fitted diffusion coefficient of the fast mode, D_0. (d)–(g) g_2 and fits at the four further Q values. Colors denote t_w, keyed along the bottom. Error bars are one-standard-deviation fit uncertainties; Section 6 gives the analysis.

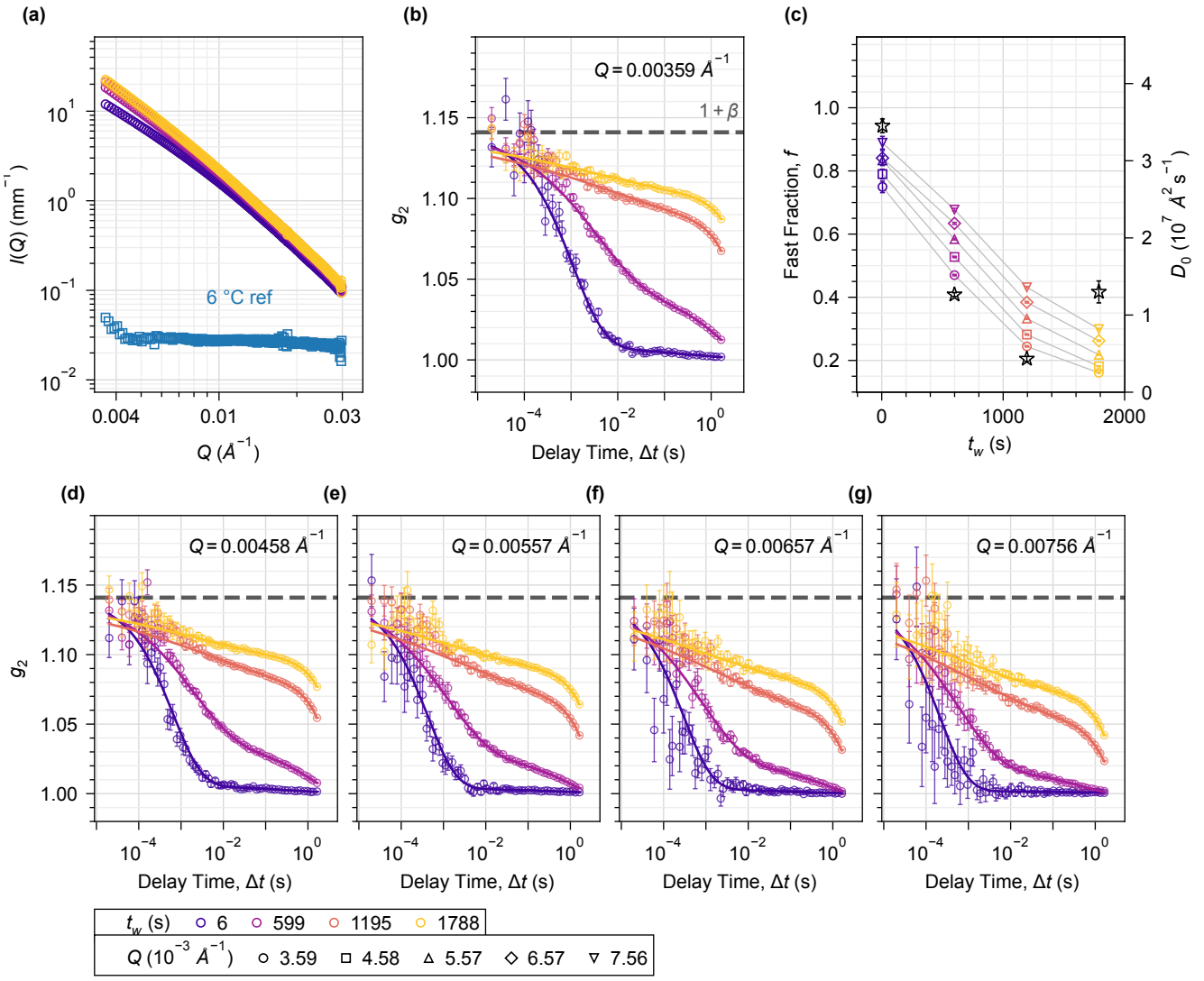


Figure S7: Loading D9 held at 28.75 °C. f at the lowest Q falls from 0.75 to 0.16 within 1800 s, increasing with Q at every t_w. (a) Background-subtracted SAXS on the absolute scale of Figure 2a of the main text: the 6 °C reference of loading D9 (blue squares), the first group at 28.75 °C and later groups. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 of the main text at $\beta = 0.141$, p_{fast} free in each group; the dashed line in (b) and (d)–(g) is $1 + \beta$, the contrast of Section 6.2. (c) Left axis, open coloured markers: the fast fraction f at the five Q values, joined by grey guide lines. Right axis, open black stars: the fitted diffusion coefficient of the fast mode, D_0. (d)–(g) g_2 and fits at the four further Q values. Colors denote t_w, keyed along the bottom. Error bars are one-standard-deviation fit uncertainties; Section 6 gives the analysis.

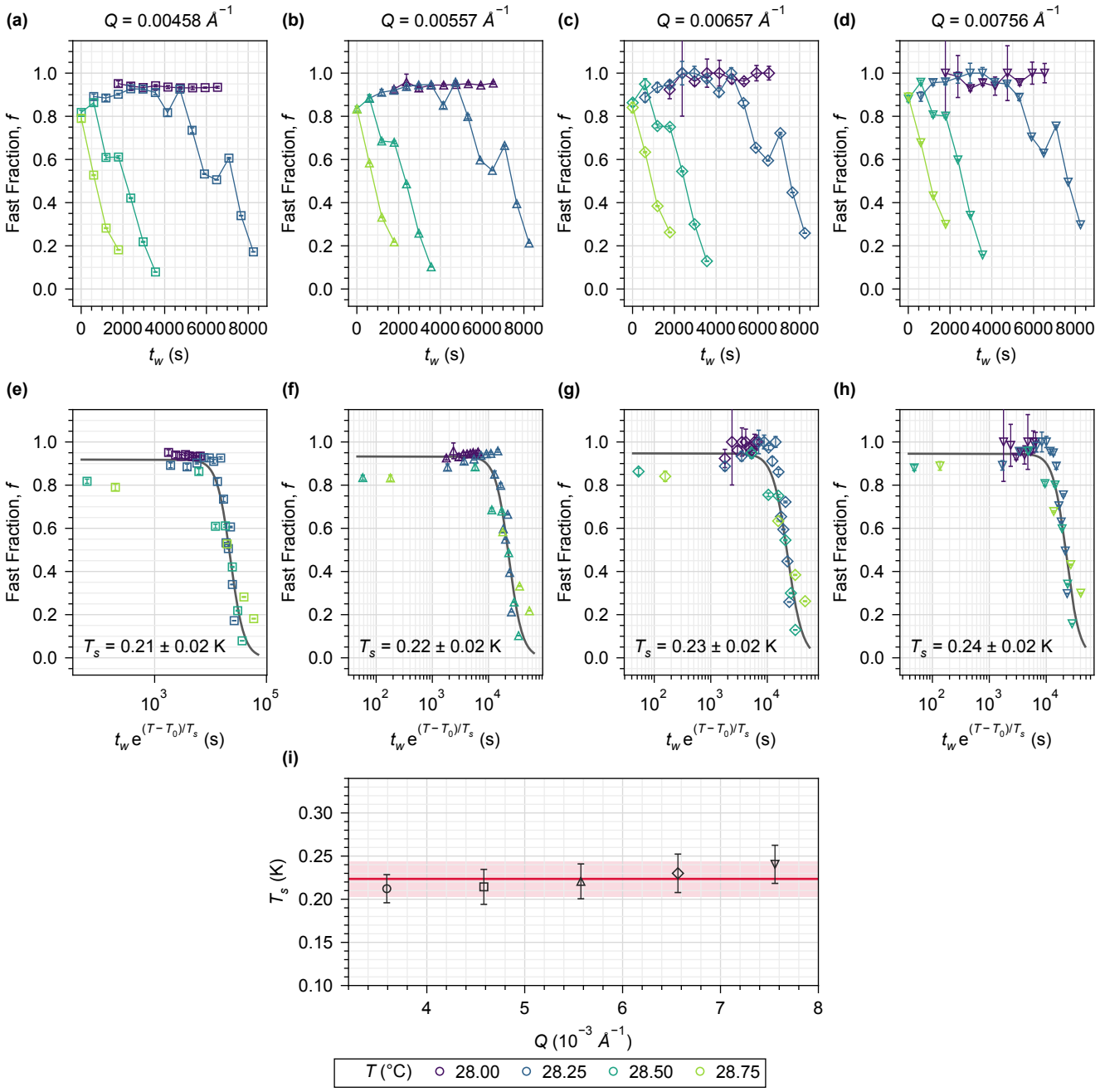


Figure S8: The collapse of Figure 3 repeated at the four higher Q bins. (a)–(d) the fast fraction against t_w; (e)–(h) the same points rescaled by $\exp[(T - T_0)/T_s]$, each panel using the T_s fitted for that bin alone and quoting it with its bootstrap uncertainty, and each with its own master curve. (i) the T_s collapsing each bin alone, against Q, with bootstrap uncertainties; the line is the mean over the five bins and the band the bootstrap uncertainty of a single-bin collapse, $T_s = 0.22 \pm 0.02$ K; the five values themselves spread by only 0.01 K, so the quoted uncertainty is the larger and more conservative of the two. Marker shape denotes Q, colour denotes temperature. The four higher bins return 0.21, 0.22, 0.23 and 0.24 K against 0.21 K at the lowest Q, so one scaling temperature describes every length scale measured to within 13%. Error bars are one-standard-deviation fit uncertainties.

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